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# SSC-I NATIONAL BOOK FOUNDATION FBISE

## CHEMISTRY TEST - CHAPTER 17: SEPARATION TECHNIQUES

| Name: | | Roll No. | | Section: |
|---|---|---|---|---|

**Total Time: 2:30 Hour**
**Time Allowed: 150 Minutes**

**Total Marks: 150**

**PAPER OBJECTIVE**
**SECTION-A (MARKS 150)**

**NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.**

**Q.1 Fill the relevant bubble for each part. All parts carry one mark.**

| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **1. Most substances we see around us are not in their pure form but are:** | Atoms | Mixtures | Elements | Compounds | ○●○○ |
| **2. A mixture contains elements and/or compounds that are not chemically:** | Separated | Heated | Present | Combined | ○○○● |
| **3. In the 1840s, people separated gold from soil by using a:** | Magnet | Pan and water | Filter paper | Distillation flask | ○●○○ |
| **4. In gold panning, the heavier _____ settled to the bottom of the pan.** | Water | Gold | Dirt | Soil | ○●○○ |
| **5. Separation of mixtures helps to obtain useful and _____ components.** | Gaseous | Harmful | Waste | Valuable | ○○○● |
| **6. Mixtures can be separated by using their _____ properties.** | Atomic only | Nuclear | Biological | Physical | ○○○● |
| **7. The method to separate a mixture depends on the _____ of its components.** | Names | Colours | Prices | Physical states | ○○○● |
| **8. Separation methods exploit differences in size, shape, mass, density or chemical:** | Cost | Colour | Name | Affinity | ○○○● |
| **9. Which is a common separation method?** | Filtration | Combustion | Polymerization | Fermentation | ●○○○ |
| **10. Another example: Which is a common separation method?** | Fermentation | Magnetic separation | Combustion | Neutralization | ○●○○ |
| **11. Yet another: Which is a common separation method?** | Fermentation | Polymerization | Combustion | Centrifugation | ○○○● |
| **12. One more: Which is a common separation method?** | Decantation | Polymerization | Combustion | Fermentation | ●○○○ |



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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **13. A solution consists of a solute and a:** | Solvent | Gas | Metal | Catalyst | ●○○○ |
| **14. In a brine solution of salt and water, the solute is:** | Air | Salt | Sand | Water | ○●○○ |
| **15. In a brine solution of salt and water, the solvent is:** | Water | Salt | Air | Sand | ●○○○ |
| **16. Evaporation is used to separate a _____ dissolved in a liquid.** | Plasma | Gas | Solid | Metal only | ○○●○ |
| **17. Evaporation is based on the fact that solids do not _____ easily, whereas liquids do.** | Vaporize | Dissolve | Freeze | Melt | ●○○○ |
| **18. In an enclosed space, a liquid will vaporize until air _____ is achieved.** | Saturation | Heating | Dryness | Cooling | ●○○○ |
| **19. Only a small fraction of molecules have enough heat energy to:** | Freeze the liquid | Vaporize the liquid | Burn the liquid | Dissolve the liquid | ○●○○ |
| **20. Increase in pressure _____ the rate of evaporation.** | Doubles | Increases | Does not affect | Decreases | ○○○● |
| **21. Increase in temperature _____ the rate of evaporation.** | Reverses | Decreases | Increases | Stops | ○○●○ |
| **22. Warm water boils faster than cold water because particles have greater:** | Size | Charge | Kinetic energy | Mass | ○○●○ |
| **23. A larger surface area _____ the rate of evaporation.** | Decreases | Stops | Increases | Reverses | ○○●○ |
| **24. Stronger intermolecular forces lead to a _____ rate of evaporation.** | Lower | Higher | Zero only | Constant | ●○○○ |
| **25. An indicator of intermolecular forces is the enthalpy of:** | Combustion | Fusion | Solution | Vaporization | ○○○● |
| **26. An increase in the flow of unsaturated air _____ the rate of evaporation.** | Stops | Decreases | Increases | Reverses | ○○●○ |
| **27. Which is NOT a factor affecting the rate of evaporation?** | Pressure | Temperature | Colour of the liquid | Surface area | ○○●○ |
| **28. Evaporation helps to separate a volatile component from a _____ one.** | Volatile | Gaseous | Liquid | Non-volatile | ○○○● |
| **29. Seawater can be evaporated to produce:** | Salt | Oil | Petrol | Gold | ●○○○ |

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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **30. Dye from ink can be separated by:** | Filtration | Sieving | Magnet | Evaporation | ○○○● |
| **31. Dairy producers use evaporation to dry lactose into:** | Gases | Powders | Crystals only | Liquids | ○●○○ |
| **32. Beverage producers make concentrates by reducing water content through:** | Evaporation | Filtration | Burning | Freezing | ●○○○ |
| **33. Distillation separates components of a _____ mixture.** | Gas only | Solid | Liquid | Metal | ○○●○ |
| **34. Distillation exploits differences in:** | Volatility | Smell | Colour | Taste | ●○○○ |
| **35. The liquid obtained from condensing vapour is called the:** | Precipitate | Distillate | Residue | Filtrate | ○●○○ |
| **36. Distillation is carried out at the solvent's:** | Melting point | Freezing point | Room temperature | Boiling point | ○○○● |
| **37. Boiling takes place when vapour pressure equals _____ pressure.** | Double | Atmospheric | Zero | Negative | ○●○○ |
| **38. In distillation, the component with the _____ boiling point vaporizes first.** | Highest | Lower | Higher | Equal | ○●○○ |
| **39. In distillation, vapour condenses in a:** | Beaker only | Water-cooled condenser | Burner | Heater | ○●○○ |
| **40. The distillate is collected in a:** | Filter | Burner | Graduated cylinder | Heater | ○○●○ |
| **41. The other liquid component remains in the:** | Graduated cylinder | Filter | Round bottom flask | Condenser | ○○●○ |
| **42. Which is a type of distillation?** | Simple distillation | Magnetic distillation | Dry distillation of wood only | Cold distillation | ●○○○ |
| **43. Another example: Which is a type of distillation?** | Magnetic distillation | Paper distillation | Fractional distillation | Cold distillation | ○○●○ |
| **44. Yet another: Which is a type of distillation?** | Steam distillation | Paper distillation | Cold distillation | Magnetic distillation | ●○○○ |
| **45. One more: Which is a type of distillation?** | Cold distillation | Vacuum distillation | Magnetic distillation | Paper distillation | ○●○○ |
| **46. Fractional distillation uses a:** | Sieve | Buchner funnel | Fractionating column | Magnet | ○○●○ |

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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **47. A fractionating column gives a temperature gradient that is cooler at the _____.** | Middle | Top | Bottom | Side | ○●○○ |
| **48. A fractionating column is hotter at the:** | Bottom | Middle | Top | Side | ●○○○ |
| **49. Fractional distillation is used to separate liquids with _____ boiling points.** | Very far | Zero | Identical | Close | ○○○● |
| **50. Steam distillation distils water together with other _____ components.** | Volatile | Solid only | Metallic | Magnetic | ●○○○ |
| **51. Steam distillation is used to obtain:** | Iron | Salt | Gold | Essential oils | ○○○● |
| **52. Vacuum distillation is performed under _____ pressure.** | Reduced | Atmospheric only | Increased | Zero only | ●○○○ |
| **53. Vacuum distillation saves time or energy and purifies compounds that are not readily:** | Frozen | Burnt | Distilled at ambient pressure | Dissolved | ○○●○ |
| **54. Humans have used distillation since about:** | 3000 BC | 300 BC | 1000 AD | 2000 AD | ●○○○ |
| **55. Distillation is used to make:** | Sugar | Salt | Distilled water | Glass | ○○●○ |
| **56. Many desalination plants use distillation to obtain drinking water from:** | Seawater | Rainwater | Lakes | Rivers | ●○○○ |
| **57. Distillation is used to refine:** | Sand | Metals | Wood | Crude oil | ○○○● |
| **58. Distillation is used to purify:** | Sand | Alcohol | Gold | Iron | ○●○○ |
| **59. Perfumes and food flavourings are obtained from herbs by:** | Distillation | Sieving | Magnets | Filtration | ●○○○ |
| **60. Air can be separated into nitrogen, oxygen and argon by _____ distillation.** | Simple | Vacuum | Steam | Cryogenic | ○○○● |
| **61. Distilled water is used in:** | Glass | Lead-acid batteries | Plastic bags | Cement | ○●○○ |
| **62. Filtration separates an _____ solid from a liquid.** | Dissolved | Gaseous | Metallic | Undissolved | ○○○● |
| **63. Filtration is based on the difference in the _____ of particles.** | Densities only | Names | Sizes | Colours | ○○●○ |

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| Questions | A | B | C | D | A B C D |
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| **64. A filter medium allows the _____ to pass through.** | Neither | Solid | Both | Liquid | ○○○● |
| **65. A filter medium blocks the _____ particles.** | Gas | Solid | Liquid | Ion | ○●○○ |
| **66. The pores of the filter medium are bigger than the _____ particles.** | Solid | Sand | Dirt | Solvent | ○○○● |
| **67. The liquid that passes through the filter is called the:** | Residue | Filtrate | Distillate | Precipitate | ○●○○ |
| **68. The solid left on the filter paper is called the:** | Solution | Residue | Filtrate | Distillate | ○●○○ |
| **69. To separate sand from water we may use a _____ cloth.** | Silk | Wool | Muslin | Nylon | ○○●○ |
| **70. Filtration by filter paper is also called _____ filtration.** | Gravity | Cold | Vacuum | Magnetic | ●○○○ |
| **71. Filter paper is a _____ barrier.** | Permeable fully | Magnetic | Semi-permeable | Impermeable | ○○●○ |
| **72. Vacuum filtration uses a:** | Buchner funnel | Condenser | Pipette | Burette | ●○○○ |
| **73. A Buchner funnel is made of:** | Porcelain | Rubber | Steel | Wood | ●○○○ |
| **74. A Buchner funnel is fitted into a side-arm:** | Round-bottom flask | Beaker | Filter flask | Test tube | ○○●○ |
| **75. A Buchner funnel is held with a _____ stopper.** | Metal | Glass | Rubber | Wooden | ○○●○ |
| **76. Filter paper for a Buchner funnel is wetted with the solvent to create a:** | Colour | Smell | Seal | Crack | ○○●○ |
| **77. Vacuum filtration is also called _____ filtration.** | Magnetic | Suction | Gravity | Cold | ○●○○ |
| **78. When making coffee, hot water is filtered through ground:** | Sugar | Rice | Salt | Coffee | ○○○● |
| **79. The filtrate in coffee making is the:** | Cup | Liquid coffee | Filter paper | Coffee grounds | ○●○○ |
| **80. A tea bag is an example of:** | Filtration | Evaporation | Crystallization | Distillation | ●○○○ |
| **81. The kidney is a biological:** | Magnet | Heater | Pump | Filter | ○○○● |

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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **82. HEPA filters are used in vacuum cleaners and air conditioners to filter out:** | Dust | Oil | Water | Gases | ●○○○ |
| **83. Belt filters are used in mining to extract:** | Salt | Sugar | Precious metals | Water | ○○●○ |
| **84. Sand and permeable rock filter water stored in the:** | Lake | Aquifer | Cloud | Ocean | ○●○○ |
| **85. Wastewater treatment plants use _____ to filter sewage.** | Crystallization | Distillation | Filtration | Evaporation | ○○●○ |
| **86. Crystallization is a technique for _____ of substances.** | Burning | Destruction | Purification | Freezing | ○○●○ |
| **87. Crystallization separates a _____ from its solution.** | Liquid | Plasma | Gas | Solid | ○○○● |
| **88. A pure crystal has a well-defined _____ structure.** | 2-D | 1-D | 3-D | Random | ○○●○ |
| **89. The principle of crystallization is based on the limited _____ of a compound.** | Density | Colour | Mass | Solubility | ○○○● |
| **90. A saturated solution is one in which undissolved solute is in equilibrium with:** | Crystals only | Dissolved solute | Air | Water vapour | ○●○○ |
| **91. In a saturated solution, dissolution and crystallization:** | Stop | Double | Reverse | Continue at equal rates | ○○○● |
| **92. To test for a saturated solution, a drop on a glass rod will go _____ on cooling.** | Blue | Cloudy | Hot | Red | ○●○○ |
| **93. During crystallization, the solution is heated until enough _____ evaporates.** | Crystals | Solute | Solvent | Filtrate | ○○●○ |
| **94. After the Bunsen burner is turned off, the saturated solution is allowed to:** | Cool | Freeze | Evaporate | Boil | ●○○○ |
| **95. On cooling, solubility _____ and crystals form.** | Increases | Decreases | Doubles | Stays same | ○●○○ |
| **96. Crystals are removed from the remaining solution by:** | Filtering | Distilling | Burning | Sieving | ●○○○ |
| **97. The crystals are then:** | Burnt | Dried | Frozen | Boiled | ○●○○ |
| **98. Crystallization is used to obtain pure _____ from seawater.** | Salt | Gold | Sand | Oil | ●○○○ |

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| Questions | A | B | C | D | A B C D |
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| **99. Crystallization is used to obtain pure _____ crystals from impure alum.** | Alum | Sand | Salt | Sugar | ●○○○ |
| **100. Crystallization is more effective than evaporation because it removes _____ impurities.** | Insoluble only | Soluble | Metallic | Gaseous | ○●○○ |
| **101. The formation of ice cubes is an example of:** | Crystallization | Evaporation | Filtration | Distillation | ●○○○ |
| **102. The formation of stalactites and stalagmites is an example of:** | Distillation | Evaporation | Filtration | Crystallization | ○○○● |
| **103. Sugar crystals forming in cooling syrup is an example of:** | Distillation | Evaporation | Crystallization | Filtration | ○○●○ |
| **104. Melting and boiling points are used to identify and check the _____ of substances.** | Name | Colour | Purity | Density only | ○○●○ |
| **105. Pure substances have a fixed:** | Melting and boiling point | Smell | Colour only | Taste | ●○○○ |
| **106. Impure substances have _____ melting and boiling points.** | Fixed | No fixed | Infinite | Zero | ○●○○ |
| **107. Pure water boils at:** | 0 °C | 100 °C | 90 °C | 50 °C | ○●○○ |
| **108. Pure water melts at:** | 100 °C | −10 °C | 0 °C | 50 °C | ○○●○ |
| **109. Impurities tend to _____ the melting point.** | Not affect | Lower | Raise | Double | ○●○○ |
| **110. Impurities tend to _____ the boiling point.** | Lower | Raise | Halve | Not affect | ○●○○ |
| **111. A pure substance melts over a _____ range.** | Infinite | Zero | Broad | Narrow | ○○○● |
| **112. An impure substance melts over a _____ range.** | Broad | Fixed | Narrow | Zero | ●○○○ |
| **113. Salt water boils at a _____ temperature than pure water.** | Lower | Higher | Same | Zero | ○●○○ |
| **114. Dark chocolate takes _____ to melt than milk chocolate.** | No time | Equal time | Shorter | Longer | ○○○● |
| **115. The change of a liquid into vapour is called:** | Freezing | Evaporation | Condensation | Melting | ○●○○ |

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| Questions | A | B | C | D | A B C D |
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| **116. The change of vapour into liquid is called:** | Condensation | Freezing | Evaporation | Melting | ●○○○ |
| **117. The unit before Separation Techniques in the book is:** | Matter | Biochemistry | Empirical Data Collection and Analysis | Hydrocarbons | ○○●○ |
| **118. The unit after Separation Techniques in the book is:** | Matter | Biochemistry | Hydrocarbons | Qualitative Analysis | ○○○● |
| **119. Which method separates iron filings from sulphur?** | Magnetic separation | Distillation | Crystallization | Filtration | ●○○○ |
| **120. Which method separates sand from water?** | Magnetic separation | Filtration | Crystallization | Distillation | ○●○○ |
| **121. Which method separates ethanol from water?** | Sieving | Magnet | Filtration | Fractional distillation | ○○○● |
| **122. Which method separates salt from salt solution?** | Filtration only | Sieving | Magnet | Evaporation | ○○○● |
| **123. Which method obtains pure water from sea water?** | Sieving | Magnet | Distillation | Filtration | ○○●○ |
| **124. Which method separates different fractions of crude oil?** | Fractional distillation | Magnet | Filtration | Sieving | ●○○○ |
| **125. Which method separates a mixture of liquids with different densities?** | Sieving | Magnet | Decantation | Crystallization | ○○●○ |
| **126. Centrifugation separates mixtures by spinning at high:** | Pressure | Humidity | Temperature | Speed | ○○○● |
| **127. Sedimentation lets suspended solids _____ to the bottom.** | Evaporate | Dissolve | Settle | Float | ○○●○ |
| **128. Decantation is the process of carefully _____ off the liquid from a settled solid.** | Boiling | Burning | Pouring | Freezing | ○○●○ |
| **129. A magnet can separate a mixture of iron and:** | Sand | Steel | Nickel | Cobalt | ●○○○ |
| **130. Which separation technique is based on difference in particle size?** | Filtration | Distillation | Evaporation | Crystallization | ●○○○ |
| **131. Which separation technique is based on difference in boiling points?** | Filtration | Distillation | Magnet | Sieving | ○●○○ |
| **132. Which separation technique is based on solubility change with temperature?** | Crystallization | Sieving | Filtration | Magnet | ●○○○ |

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| Questions | A | B | C | D | A B C D |
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| **133. Which separation technique uses the volatility of one component?** | Magnet | Filtration | Sieving | Evaporation | ○○○● |
| **134. In crystallization, the saturated solution forms crystals when it:** | Evaporates fully | Boils | Cools | Freezes only | ○○●○ |
| **135. A kitchen strainer is an example of:** | Crystallization | Distillation | Evaporation | Filtration | ○○○● |
| **136. Salt pans used to get salt from sea water apply:** | Magnet | Filtration only | Distillation only | Evaporation and crystallization | ○○○● |
| **137. Drying clothes in the sun is an example of:** | Filtration | Distillation | Evaporation | Crystallization | ○○●○ |
| **138. The apparatus used in distillation to cool the vapours is the:** | Burner | Condenser | Flask | Funnel | ○●○○ |
| **139. The apparatus used in filtration to hold filter paper is a:** | Funnel | Pipette | Burette | Condenser | ●○○○ |
| **140. In simple distillation, a thermometer is placed near the:** | Funnel | Burner | Side arm | Bottom of flask | ○○●○ |
| **141. Which gas leaves at the top of a fractionating column of crude oil?** | Kerosene | Bitumen | Diesel | Petroleum gases | ○○○● |
| **142. Bitumen collects at the _____ of the fractionating column.** | Middle | Top | Side | Bottom | ○○○● |
| **143. Heat is necessary for _____ of a solution before crystallization.** | Concentrating | Drying | Cooling | Filtering | ●○○○ |
| **144. Pure crystals can be obtained from a solution by:** | Boiling to dryness | Freezing | Cooling a hot saturated solution | Filtering a cold solution | ○○●○ |
| **145. Which mixture can be separated by decantation?** | Salt and water | Ink | Alcohol and water | Sand and water | ○○○● |
| **146. Which mixture can be separated by evaporation?** | Iron and sulphur | Oil and water | Salt and water | Sand and water | ○○●○ |
| **147. Which mixture can be separated by a separating funnel?** | Sand and salt | Oil and water | Iron and sulphur | Salt and water | ○●○○ |
| **148. Chocolate melts easily in the mouth, which shows milk chocolate has a _____ melting range.** | Fixed | Infinite | Lower | Higher | ○○●○ |
| **149. The purity of a solid can be checked by its:** | Colour only | Melting point | Taste | Smell | ○●○○ |

| Questions | A | B | C | D | A B C D |
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| **150. The purity of a liquid can be checked by its:** | Density only | Smell | Taste | Boiling point | ○○○● |

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